Pimpri Chinchwad Education Trust's
Pimpri Chinchwad College of Engineering (PCCoE), Pune
Department of Computer Engineering — Research Methodology

RESEARCH PROPOSAL

Assignment No. 5

BhashaBridge: A Lightweight, Camera-Only Hybrid 1D-CNN–BiLSTM–Multi-Head Temporal Attention Framework with Wrist Root-Anchor Normalization for Continuous Real-Time Marathi Sign Language Translation

Team Member	PRN	Role
Adwait Kamble	123B1D002	Group 13
Yadnesh Patil	123B1D034	Group 13
Abhay Bormale	123B1D057	Group 13
Atharva Zope	123B1D067	Group 13

Project Guide: Prof. A. S. Birajdar

Final-Year B.Tech Capstone Project • October 2026

1. Proposed Research Title and Abstract

Title

BhashaBridge: A Lightweight, Camera-Only Hybrid 1D-CNN–BiLSTM–Multi-Head Temporal Attention Framework with Wrist Root-Anchor Normalization for Continuous Real-Time Marathi Sign Language Translation

Abstract

Marathi Sign Language (MSL) is the primary mode of communication for a large Deaf and speech-impaired community in Maharashtra, yet automatic translation tools for it are almost non-existent. Existing sign language recognition research concentrates on American and generalized Indian Sign Language and mostly treats recognition as the classification of static alphabets or isolated words captured under controlled conditions. Pixel-based 3D-CNN and Transformer pipelines are also computationally expensive, sensitive to lighting and background, and unsuited to CPU-only devices. Closed-set classifiers without a "no-sign" class additionally emit confident but spurious tokens while the signer is resting, a failure we term temporal hallucination. This proposal presents BhashaBridge, a camera-only system for continuous, real-time MSL translation. MediaPipe Holistic converts each webcam frame into a 126-dimensional skeletal keypoint vector, so no image pixels are retained. Each hand is normalized relative to its wrist (root anchor) and scaled by palm length, giving invariance to signer position and camera distance. A 30-frame rolling buffer feeds a hybrid classifier of 1D-CNN blocks for local motion features, a Bidirectional LSTM for temporal context and Multi-Head Temporal Attention for frame weighting. An explicit Idle class, a confidence threshold, multi-window debouncing and a cooldown lock suppress false activations. Recognized signs are composed into Marathi text and spoken using gTTS. A custom dataset of about 2,000 continuous Subject–Object–Verb (SOV) sentences will be recorded, informed by field research at Chinchwad Mook Badhir Vidyalaya. Evaluation uses Word Error Rate, BLEU, accuracy, false activations per minute and latency against vision-based baselines. The expected outcome is a lightweight, privacy-preserving assistive tool for hospitals, schools and public offices.

Keywords: Marathi Sign Language; Continuous Sign Language Recognition; MediaPipe Holistic; 1D-CNN; BiLSTM; Temporal Attention; Idle-State Rejection; Assistive Technology.

2. Introduction and Background

2.1 The Deaf and Hard-of-Hearing Community in Maharashtra

Sign languages are complete natural languages with their own phonology, morphology and grammar; they are not signed renderings of spoken languages [8]. For many Deaf and speech-impaired people in Maharashtra, sign language is the first and preferred means of communication. India's Rights of Persons with Disabilities Act, 2016 recognizes sign language as a mode of communication. However, qualified interpreters are scarce, so Deaf individuals often depend on family members or on writing when they visit hospitals, banks, government offices and schools. Hearing non-signers rarely know any sign language, and many Deaf users have limited fluency in written Marathi. The result is a two-way communication barrier that restricts access to healthcare, education, employment and public services.

2.2 Marathi Sign Language

MSL is used by Marathi-speaking Deaf communities and is related to, but not identical with, Indian Sign Language (ISL). It has regional lexical variation and a sentence structure that follows Subject–Object–Verb (SOV) order. Spoken Marathi, by contrast, is morphologically rich, with case markers, verb agreement and tense inflection. Translating signs into fluent Marathi therefore requires more than word-by-word substitution.

2.3 AI and Deep Learning in Sign Language Recognition

Sign language recognition (SLR) has moved through three broad stages:

1. **Handcrafted vision pipelines** (colour-space segmentation, edge detection, template matching), which work mainly in controlled scenes.
2. **Deep appearance models** (2D/3D-CNNs, CNN–LSTM hybrids, Transformers) that learn directly from RGB frames and give strong benchmark results at high computational cost [7], [8].
3. **Landmark (skeleton)-based models**, where on-device trackers such as MediaPipe [1], [2] reduce each frame to a compact set of 3D keypoints. This lowers computation, removes dependence on appearance, and keeps faces and backgrounds out of storage.

Sequence-modelling components that suit landmark streams include LSTM [3], bidirectional recurrent networks [4], self-attention [5] and temporal convolutions [6]. Evidence for combining all three in one hybrid for continuous regional sign language remains limited, which motivates this work.

3. Problem Identification and Research Questions

3.1 Problem Statement

There is no practical, real-time, sentence-level MSL translator that runs on ordinary hardware. Existing systems typically (i) target ASL or ISL, (ii) classify static alphabets or isolated words, (iii) rely on heavy GPU-bound pixel models, and (iv) degrade in real conversation. Real signing contains co-articulation between signs, no clear word boundaries, and frequent pauses and self-adaptors (resting hands, scratching, posture changes). A closed-set softmax classifier must assign probability to some sign class even for such non-signing input, so it emits confident but meaningless tokens (temporal hallucination).

3.2 Research Questions

ID	Research Question
RQ1	Can a hybrid 1D-CNN + BiLSTM + Multi-Head Temporal Attention model trained on wrist-normalized MediaPipe keypoints accurately recognize continuous, SOV-structured MSL sentences?
RQ2	How much does wrist root-anchor normalization improve signer-independent and camera-independent generalization compared with raw image-space landmark coordinates?
RQ3	Can an explicit Idle class combined with confidence thresholding, multi-window debouncing and cooldown locking reduce false activations per minute without unacceptable loss of responsiveness?
RQ4	Does the proposed lightweight model achieve lower inference latency on CPU-only hardware than RGB-based 3D-CNN baselines, at comparable or better translation quality?

TABLE I. RESEARCH QUESTIONS

4. Literature Review and Research Gap

4.1 Appearance-Based and Static Recognition

A large share of Indian SLR work classifies static alphabets, digits and postures from single RGB frames using shallow CNNs, transfer-learned backbones (VGG, ResNet, MobileNet) and, more recently, Vision Transformers. These reach high accuracy on closed datasets. Reported weaknesses include sensitivity to illumination, skin tone, background clutter and camera framing, plus an inability to represent motion trajectories or co-articulation [8]. Marathi-specific work is largely restricted to static Devanagari alphabet recognition.

4.2 Continuous Recognition and Translation

Continuous SLR on benchmarks such as RWTH-PHOENIX-Weather uses CNN–BiLSTM encoders trained with Connectionist Temporal Classification and, later, Transformer-based joint recognition and translation [7]. These systems model a full unsegmented video offline. They commonly rely on 3D-CNN or large 2D backbones that need GPU acceleration, which makes them poorly suited to low-latency, CPU-only use.

4.3 Skeleton-Based Approaches

MediaPipe Hands and Holistic provide real-time 21-point hand landmarks on commodity devices [1], [2]. Landmark-based SLR systems use LSTMs, graph convolutions or Transformers, mostly on isolated or pre-segmented dynamic signs. Three limitations recur across this body of work:

- Raw landmark coordinates are expressed in image space, so they shift with signer position and distance from the camera.
- Evaluation is usually on pre-segmented clips, so idle and transition behaviour in live streams is rarely measured.
- Unidirectional recurrent units lack look-ahead context within a window, and pure Transformers lack local inductive bias on short windows.

4.4 Datasets

Large benchmarks exist for other languages, such as WLASL for ASL and INCLUDE [9] for ISL words, but publicly available MSL resources are small, mostly static, and recorded by few signers. In the literature reviewed by the team, no peer-reviewed, multi-signer, continuous, sentence-level MSL benchmark was found.

4.5 Research Gap

Based on the preceding review, the gap is defined along four dimensions:

4. **Language gap:** Marathi Sign Language lacks continuous, sentence-level datasets and translators.
5. **Temporal gap:** most systems treat signs as static or isolated rather than as continuous, co-articulated streams.
6. **Robustness gap:** landmark pipelines seldom apply geometric normalization for translation and scale invariance, and seldom report idle-state rejection or false-activation metrics.
7. **Deployability gap:** pixel-heavy 3D-CNN/Transformer models are impractical on commodity CPUs, while cloud inference adds latency and privacy risk.

BhashaBridge addresses these gaps jointly through a zero-pixel, normalized-landmark hybrid classifier and an inference-control layer, trained on a purpose-built SOV MSL corpus.

5. Research Objectives

- **O1 (Dataset):** Design and record a custom continuous MSL dataset of about 2,000 SOV-structured sentences from multiple signers, with sign-level, sentence-level and Idle/transition annotations and a signer-independent train/validation/test partition.
- **O2 (Feature representation):** Implement a MediaPipe Holistic extraction stage and a wrist root-anchor normalization scheme (translation- and scale-invariant), and quantify its effect through ablation.
- **O3 (Model):** Develop and train a hybrid 1D-CNN + BiLSTM + Multi-Head Temporal Attention classifier with an explicit (N + 1)-class Idle output for continuous MSL.
- **O4 (Inference optimization):** Build a rolling-buffer inference engine with a confidence gate, k-window debounce and cooldown lock that minimizes false activations per minute while keeping latency within real-time limits on CPU-only hardware.
- **O5 (System integration and evaluation):** Integrate recognition with Marathi sentence composition, on-screen text, gTTS audio output and a GUI, and evaluate with WER, BLEU, accuracy, latency and usability testing against RGB-based baselines.

6. Hypotheses Proposed

Primary hypothesis (H1): A hybrid 1D-CNN + BiLSTM + Multi-Head Temporal Attention model using wrist-normalized MediaPipe skeletal keypoints will achieve lower CPU inference latency and equal or higher recognition and translation quality for continuous MSL than standard RGB-based 3D-CNN baselines.

Supporting hypotheses:

ID	Null hypothesis (H_0)	Alternative hypothesis (H_a)
H1a	Wrist root-anchor normalization has no effect on signer-independent accuracy.	Normalization significantly improves signer-independent accuracy and accuracy under changed camera distance and framing.
H1b	Adding temporal attention to 1D-CNN + BiLSTM does not change WER.	Temporal attention significantly reduces WER.
H1c	An Idle class with threshold, debounce and cooldown does not reduce false activations per minute (FAPM).	The full inference-control stack significantly reduces FAPM on idle and ambient-motion recordings.
H1d	Landmark-based inference is not faster than RGB 3D-CNN inference on CPU.	Landmark-based inference achieves significantly lower per-window latency.

TABLE II. NULL AND ALTERNATIVE HYPOTHESES

Hypotheses will be tested with paired bootstrap resampling over test sentences (95% confidence intervals) and McNemar's test for paired classification outcomes, with significance at $\alpha = 0.05$.

7. Proposed Methodology (Algorithm / Technique)

7.1 Overview

BhashaBridge is organized in three stages: **Stage A** (data acquisition and preprocessing), **Stage B** (hybrid deep-learning classification) and **Stage C** (post-processing and output). The complete architecture is shown in Fig. 1, the live inference loop in Fig. 2 and the overall research workflow in Fig. 3.

7.2 Stage A: Data Capture and Skeleton Extraction

8. **Capture:** webcam video at 30 FPS, 640 × 480.
9. **MediaPipe Holistic extraction [1], [2]:** for each frame the left- and right-hand streams (21 landmarks × 3 coordinates each) are retained, giving a 126-dimensional vector. The frame is then discarded, so no pixels are stored or transmitted (the *zero-pixel* property).
10. **Missing-hand zero-fill:** if a hand is not detected, its 63 values are set to zero so that the vector shape stays fixed at 126.
11. **Wrist root-anchor normalization:** for frame t, hand h and landmark j, each coordinate is expressed relative to the wrist (landmark 0) and divided by the palm length (distance from the wrist to the middle-finger MCP knuckle, landmark 9):

$$\tilde{p}(t,h,j) = \left(p(t,h,j) - p(t,h,0) \right) / \left(\| p(t,h,9) - p(t,h,0) \|_2 + \varepsilon \right) \quad (1)$$

Here $\varepsilon = 10^{-6}$ prevents division by zero. The result is invariant to in-plane translation and to scale changes caused by camera distance, and a zero-filled hand remains zero. Because anchoring each hand independently discards inter-hand distance, an optional 3-D inter-wrist displacement vector $\Delta p(t) = p(t,R,0) - p(t,L,0)$ can be appended (ablation variant).

12. **Rolling buffer:** a 30-frame FIFO window (about 1 s, step of 1 frame) yields the tensor $\mathbf{X} \in \mathbb{R}^{30 \times 126}$ on every new frame. Window lengths of 30, 45 and 60 frames will be compared, since longer windows may suit multi-phase signs.

7.3 Stage B: Hybrid Deep-Learning Classifier

(a) 1D-CNN (local motion features) [6]. Two stacked Conv1D blocks (64 and 128 filters, kernel size 3, padding "same"), each followed by BatchNorm, ReLU and Dropout (0.3), convolve along the time axis. They extract short-range velocity and acceleration patterns in parallel. Output: $C \in \mathbb{R}^{30 \times 128}$.

(b) Bidirectional LSTM (temporal sequence learning) [3], [4]. 128 units per direction with sequence return. Forward and backward states are concatenated, giving $H \in \mathbb{R}^{30 \times 256}$. This provides onset and offset context within the window, which a unidirectional LSTM lacks.

$$c_t = \mathrm{ReLU}\left(\mathrm{BN}\left(W_c * \tilde{x}_t + b_c \right) \right); \quad h_t = \left[\mathrm{LSTM}_{fwd}(c_t) ; \mathrm{LSTM}_{bwd}(c_t) \right] \quad (2)$$

(c) Multi-Head Temporal Attention [5]. $M = 4$ heads with $d_k = 64$ per head compute scaled dot-product attention over H, followed by a residual connection, Layer Normalization and global temporal average pooling:

$$A_m = \mathrm{softmax}\left(Q_m K_m^T / \sqrt{d_k} \right) V_m ; \quad \hat{y} = \mathrm{softmax}\left(W_o \cdot \mathrm{Pool}\left(\mathrm{LN}\left(H + \mathrm{Concat}(A_1..A_4) W_p \right) \right) + b_o \right) \quad (3)$$

Attention assigns higher weight to the core stroke frames and lower weight to boundary, transition and rest frames.

(d) Classifier. A Dense layer maps 256 → (N + 1), where N is the number of MSL sign classes and class 0 is the explicit **Idle** class (resting hands, empty frames, posture adjustment, self-adaptors, inter-sign transitions). The reference configuration has roughly 0.6–0.8 million trainable parameters, depending on N; the exact count will be reported after implementation.

Training. Categorical cross-entropy (optionally with label smoothing), Adam optimizer, early stopping and learning-rate scheduling. Landmark-space augmentation includes small rotation, scaling, jitter, temporal speed variation and random frame dropout to mimic tracker failure. Class weighting handles Idle-class imbalance. Training and ablation runs will use the college's GPU facility.

7.4 Stage C: Inference Control and Output

A prediction is accepted only when all of the following hold:

13. the top class is **not Idle**;
14. the maximum posterior exceeds $\tau = 0.85$ (to be calibrated on the validation set using reliability diagrams and Expected Calibration Error);
15. the same class holds for **k = 3 consecutive windows** (debounce);
16. the system is **not in a cooldown lock** ($T_{cool} = 15$ frames ≈ 0.5 s), which prevents duplicate emission while a sign is held.

$$E_w = \mathbb{1}(c_w \neq 0) \cdot \mathbb{1}(p_w > \tau) \cdot \prod_{i=0}^{k-1} \mathbb{1}(c_{(w-i)} = c_w) \cdot \mathbb{1}(w \notin T_{cool}) \quad (4)$$

On Idle, the debounce counter resets and the cooldown is released. Emitted sign tokens are appended to a sentence buffer. Because SOV gloss order differs from natural inflected Marathi, a rule-based composition module (a gloss-to-Marathi template and inflection table covering the dataset vocabulary) generates the Devanagari sentence. It is displayed in the GUI and synthesized with **gTTS** (language code *mr*), played asynchronously so recognition is not blocked. Because gTTS needs internet access, an offline fallback (e.g., eSpeak-NG) will be evaluated for low-connectivity settings.

7.5 System Architecture Diagram

The complete architecture is presented in Fig. 1 (IEEE format – Black and White).

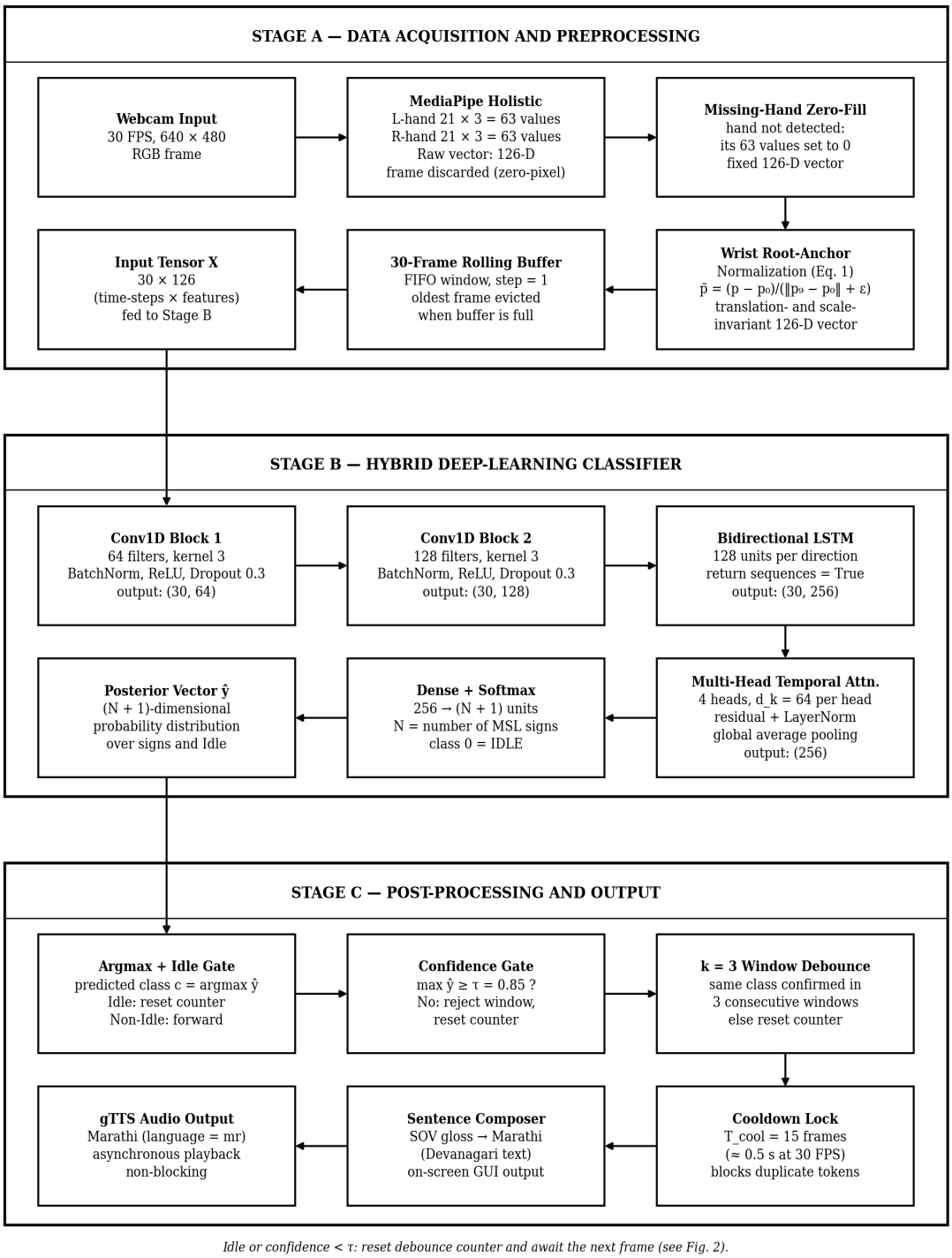


Fig. 1. BhashaBridge system architecture: Stage A (acquisition and preprocessing), Stage B (hybrid 1D-CNN + BiLSTM + temporal-attention classifier) and Stage C (post-processing and output). *IEEE format – Black and White.*

7.6 Live Inference Flowchart

The continuous inference loop, including Idle rejection, confidence gating, debouncing and cooldown, is shown in Fig. 2 (IEEE format – Black and White).

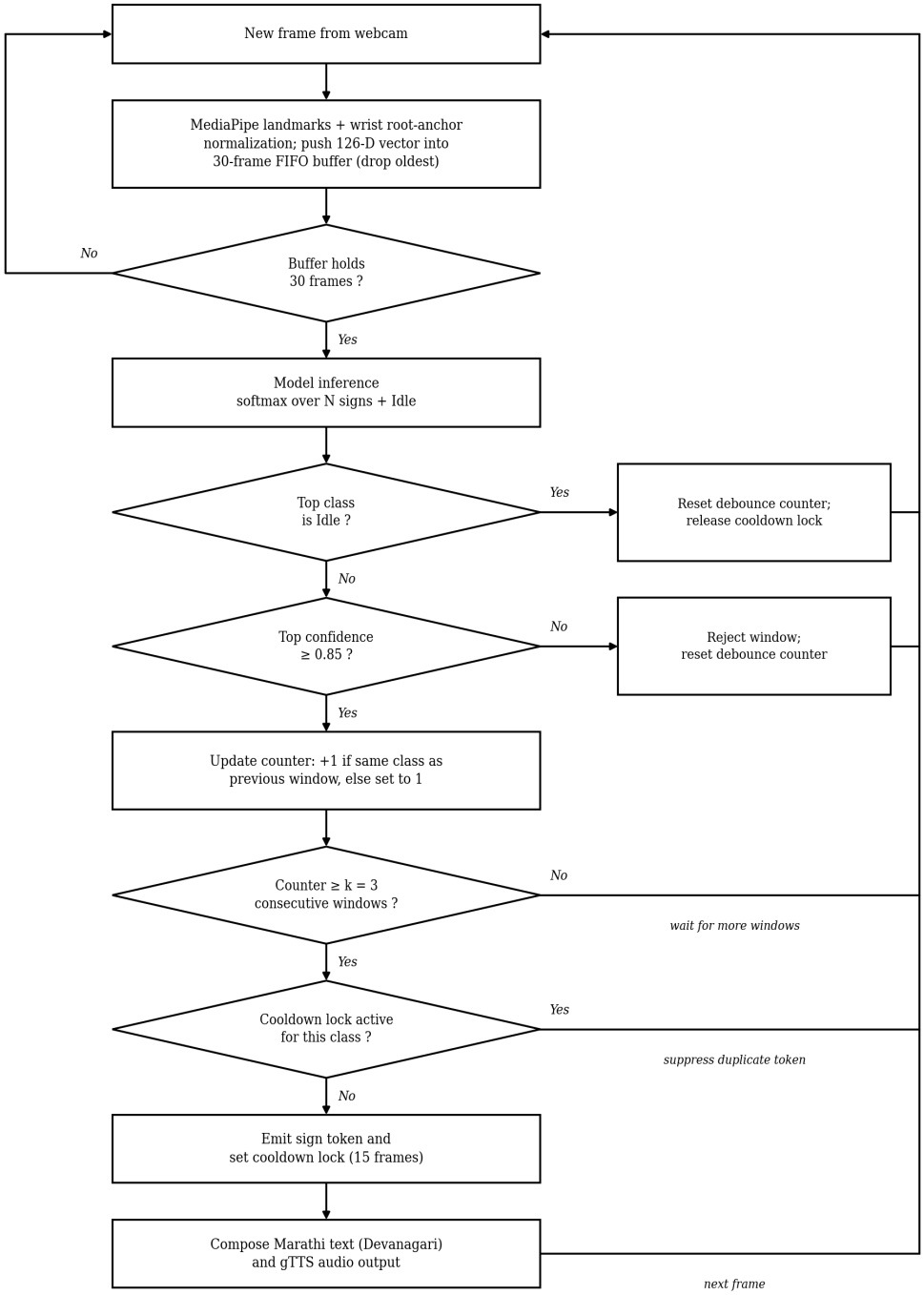


Fig. 2. Continuous live inference workflow with rolling buffer, Idle gate, 0.85 confidence gate, k = 3 debounce and cooldown lock. *IEEE format – Black and White.*

7.7 Research Methodology Flowchart

The overall research workflow, grouped into five phases with a training-validation feedback loop, is shown in Fig. 3 (IEEE format – Black and White).

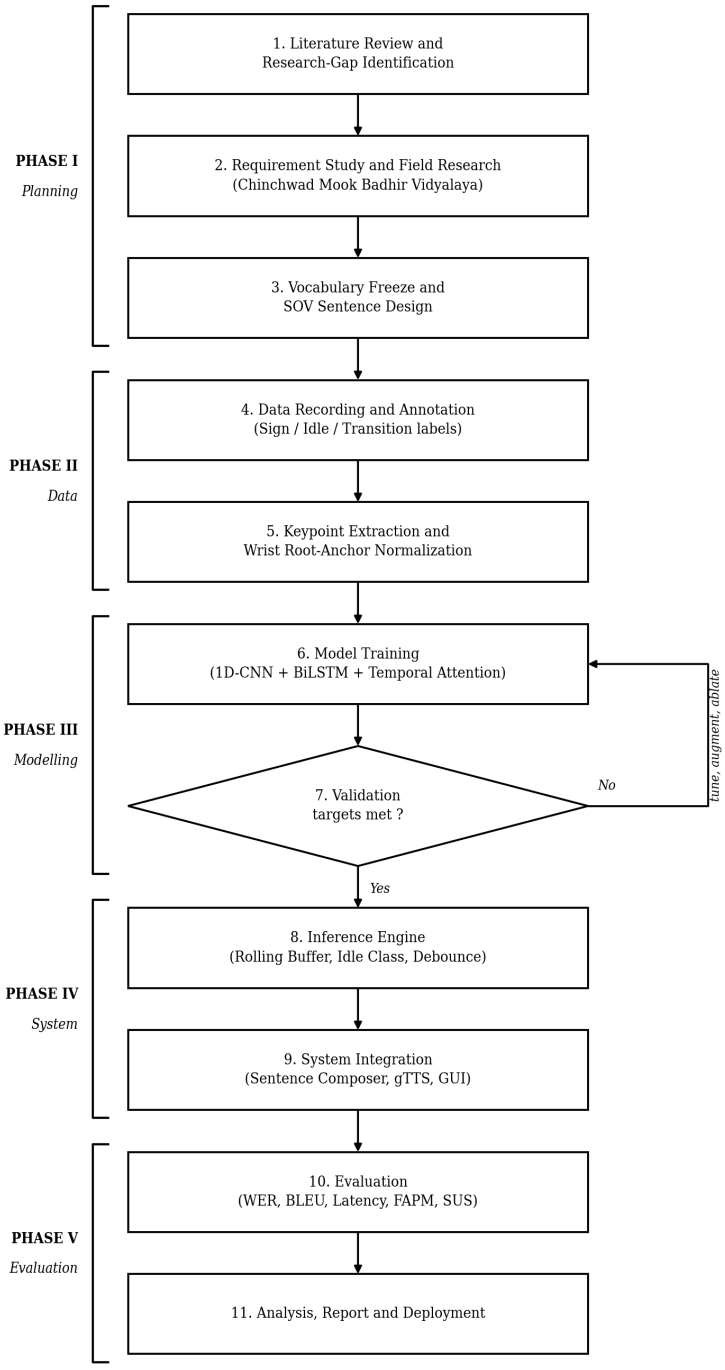


Fig. 3. Overall research methodology flowchart. *IEEE format – Black and White.*

8. Dataset, Sampling and Data Collection

8.1 Field Study Context

Requirements gathering and field research were conducted at **Chinchwad Mook Badhir Vidyalaya**, a school for deaf and speech-impaired students. These visits informed the choice of everyday-communication domains, the sign variants in local use and the practical recording constraints. Teachers or interpreters from the school will serve as linguistic validators for the gloss and translation pairs.

8.2 Ethics and Consent

Informed consent will be obtained from adult participants, and from parents or guardians (with the participant's assent) for minors, together with authorization from the school. Only landmark coordinates will be stored in the released dataset, with no identifiable video retained. Participants will be assigned anonymous IDs, participation will be voluntary, and participants may withdraw at any time.

8.3 SOV Grammar Mapping and Corpus Design

MSL sentences follow SOV order. For each target Marathi sentence a validated gloss sequence is defined. For example, a sentence meaning "I want to meet the doctor today" maps to a gloss order such as *I – TODAY – HOSPITAL – DOCTOR – MEET – WANT* (final glosses to be validated by native signers). The corpus is designed as follows:

- **Target size:** about 2,000 continuous sentences.
- **Vocabulary:** a fixed lexicon of roughly 300–500 lexical signs, frozen after the field study.
- **Domains:** healthcare, education, banking and public services, daily needs and emergencies.
- **Sentence length:** 3–7 signs per sentence with a balanced distribution.
- **Coverage:** every vocabulary sign appears in several sentence contexts to support co-articulation learning.

8.4 Recording Protocol

Parameter	Specification
Signers	10–12 signers, including native Deaf signers and trained interpreters
Capture device	Laptop webcams plus one external 1080p webcam, used at 640 × 480
Sampling rate	30 frames per second
Variation	Multiple rooms, lighting levels, camera distances, heights and angles, and clothing
Sessions	Several sessions per signer on different days
Stored data	Per-frame 126-D landmark arrays (optional pose landmarks)
Classes	N MSL sign classes plus an explicit Idle class

TABLE III. RECORDING PROTOCOL

Each signer records a subset of sentences, repeated several times. Dedicated recordings capture the **Idle/Transition class**: hands resting in the lap or at the side, hands out of frame, scratching, adjusting glasses or clothing, posture shifts and listening pauses.

8.5 Annotation

Annotation uses sentence-level labels (gloss sequence and Marathi translation) and sign-level temporal boundaries, with transition and idle segments marked. At least two annotators will label independently and a native-signer validator will resolve disagreements. Inter-annotator agreement (Cohen's κ) will be reported.

8.6 Sampling and Data Partitioning

- **Sampling:** purposive sampling, selecting signers of different ages, fluency levels and regional variants.
- **Partition:** a **signer-independent** split (about 70% train / 15% validation / 15% test by signer, not by clip), to avoid inflating accuracy through signer-specific hand geometry. An additional held-out camera/room condition will test cross-environment generalization.
- **Windowing:** training windows are created with a sliding 30-frame window and labelled by the sign occupying the dominant temporal portion, or Idle where no sign is active.

The collection pipeline is summarized in Fig. 4 (IEEE format – Black and White).

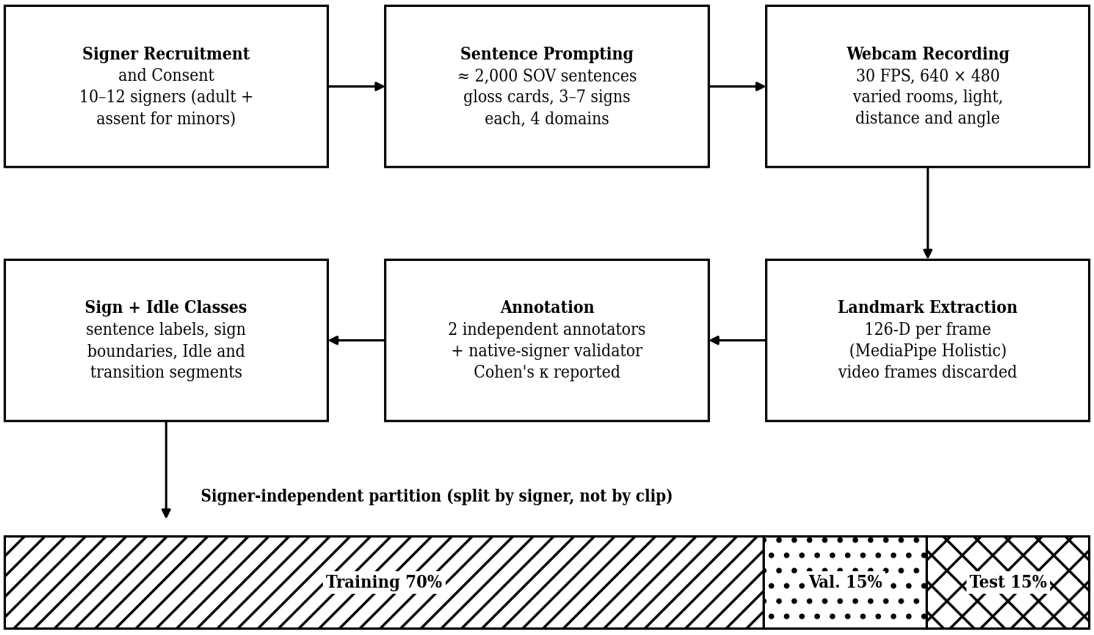


Fig. 4. Dataset collection, annotation and signer-independent partition pipeline. *IEEE format – Black and White.*

9. Data Analysis and Evaluation Plan

9.1 Evaluation Metrics

Metric	Definition / Use
Accuracy (%)	Window-level and sign-level accuracy, plus macro-F1 to account for class imbalance.
Word Error Rate (WER)	WER = (S + D + I) / N over emitted sign sequences, where S, D and I are substitutions, deletions and insertions and N is the reference length.
BLEU	Corpus-level BLEU [10] between generated and reference Marathi sentences, reported as BLEU-4 with lower-order n-gram scores.
Latency (ms)	Per-window inference time and end-to-end delay (frame capture to text to audio start) on CPU-only hardware; mean, median and 95th percentile.
Throughput (FPS)	Sustained frames per second on CPU.
FAPM	False Activations Per Minute: spurious tokens emitted per minute on long idle and ambient-motion recordings.
ECE	Expected Calibration Error of the softmax confidence, used to calibrate τ.
Model footprint	Parameter count, model size (MB) and FLOPs.
Usability (SUS)	System Usability Scale scores from Deaf and hearing participants.

TABLE IV. EVALUATION METRICS

$$WER = (S + D + I) / N \times 100\ \% \quad (5)$$

9.2 Experimental Design

Baselines: (i) an RGB-based 3D-CNN (I3D-style) pipeline; (ii) a 2D-CNN + BiLSTM RGB model; (iii) an LSTM on raw (unnormalized) landmarks; (iv) a Transformer encoder on landmarks. **Ablation study:** each variant below removes or alters one design element.

Variant	Purpose
Raw coordinates vs. wrist-normalized	Tests H1a
1D-CNN + BiLSTM, with vs. without attention	Tests H1b
BiLSTM vs. unidirectional LSTM	Isolates bidirectional context
With vs. without Idle class	Tests H1c
With/without threshold, debounce and cooldown	Tests H1c
Window length 30 / 45 / 60 frames	Selects buffer size
With vs. without inter-wrist displacement	Tests loss of inter-hand information

TABLE V. ABLATION STUDY PLAN

A **threshold sweep** (τ from 0.5 to 0.95, k from 1 to 5) will plot the trade-off between FAPM and recognition delay. **Robustness tests** will vary signers, webcams, distances, lighting (dim/normal/bright) and moderate background motion. **Latency benchmarking** will be run on a CPU-only laptop against the RGB baseline.

9.3 Statistical Analysis

Results will be reported as mean ± standard deviation across multiple random seeds, with 95% bootstrap confidence intervals. Paired bootstrap resampling and McNemar's test will compare models, with Holm or Bonferroni correction for multiple comparisons. Per-class confusion matrices will identify confusable signs, and usability feedback will be summarized descriptively.

10. Expected Results, Significance and Contribution

10.1 Expected Results

The following are design targets to be verified experimentally, not claims:

- Lower per-window CPU latency than RGB 3D-CNN baselines, reported as a measured speed-up, with real-time operation (≥ 25–30 FPS) on a CPU-only machine.
- Measurably higher signer-independent accuracy with root-anchor normalization than with raw coordinates.
- Substantially reduced FAPM when the Idle class and inference-control stack are applied.
- Competitive WER and BLEU on the custom MSL corpus, with honest reporting of its vocabulary-limited scope.

10.2 Societal Significance

A working MSL translator can help Deaf and speech-impaired Marathi speakers communicate with hearing non-signers in hospitals, schools, banks and public offices, reducing dependence on scarce interpreters. Producing Marathi text and speech lets the signer be understood directly in the local language. Because the system needs only a webcam and stores no images, it is affordable and privacy-preserving.

10.3 Technical Contributions

17. A **zero-pixel, wrist root-anchored** representation for translation- and scale-invariant MSL recognition.
18. A **lightweight hybrid 1D-CNN + BiLSTM + Multi-Head Temporal Attention** architecture with an explicit Idle output.
19. A **calibrated inference-control scheme** (threshold, debounce, cooldown) with a reported FAPM metric.
20. A **custom continuous SOV MSL corpus** with documented protocols and signer-independent splits, to be released subject to consent and institutional approval.
21. A reproducible **evaluation protocol** and CPU-latency benchmark for regional sign language translation.

10.4 Hardware Deployability

Because input dimensionality falls from tens of thousands of pixel values per frame to 126 landmark values, the model is expected to run on ordinary laptops and, after quantization (e.g., ONNX Runtime or TensorFlow Lite INT8), potentially on budget smartphones or small kiosk devices. This is proposed future work, and feasibility will be measured rather than assumed.

11. Scope, Assumptions and Limitations

11.1 Scope

- Limited to Marathi Sign Language and a fixed vocabulary of roughly 300–500 signs within the chosen domains.
- Camera-only input using a standard webcam, with no gloves or depth sensors.
- Sign-to-Marathi text and speech (one direction). Speech-to-sign avatar synthesis is future work.
- Single-signer operation, with the signer facing the camera.

11.2 Assumptions

- Reasonably well-lit environments with the signer’s upper body and both hands in view.
- Signers use the standardized lexicon adopted in the dataset.
- Adequate webcam quality (at least 480p and 30 FPS) and stable positioning.
- MediaPipe provides sufficiently reliable landmarks under the stated conditions.
- Native-signer validation of glosses and translations is available.

11.3 Limitations

- **Occlusion:** hand-to-hand overlap, hand-to-face contact and fast motion can cause landmark dropout or jitter.
- **Background noise:** other people moving in view may confuse tracking, since MediaPipe may lock onto the wrong person.
- **Non-manual markers:** a hands-only stream omits facial expression, mouth patterns and head movement, so question forms and negation may be ambiguous. Face and upper-body landmarks are future work.
- **Vocabulary:** a closed-set classifier cannot recognize unseen signs and does not handle fingerspelling beyond the included vocabulary.
- **Dialects:** regional lexical differences across Maharashtra are only partly covered.

- **gTTS dependency:** online synthesis needs internet access, and an offline fallback may have lower voice quality.
- **Window size:** a fixed 30-frame buffer may cut very long signs, which is why window length is ablated.
- **Sentence composition:** the rule-based gloss-to-Marathi step covers only the dataset's structures; a learned translation module is future work.

12. Work Plan and Timeline

The plan spans **10 months**, from literature review and dataset collection to model training, GUI integration and final deployment. Filled (black) cells mark the active months of each activity.

Activity	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10
1. Literature review and gap analysis	■	■								
2. Field study and requirements	■	■								
3. Vocabulary and SOV design		■	■							
4. Dataset recording			■	■	■					
5. Annotation and quality checks				■	■	■				
6. Extraction and normalization			■	■						
7. Baseline model implementation				■	■					
8. Hybrid model training and ablation					■	■	■			
9. Inference engine and tuning						■	■			
10. Sentence composer, gTTS and GUI							■	■		
11. Evaluation and user testing								■	■	
12. Optimization and deployment									■	■
13. Report and paper writing									■	■

Fig. 5. Project timeline (Gantt chart). *IEEE format – Black and White.*

Milestones

Month	Milestone
M2	Literature review complete, research gap finalized, field-study report
M3	Vocabulary and SOV sentence list frozen
M5	Dataset recording complete
M6	Annotation finished and signer-independent splits fixed
M7	Hybrid model trained and baselines benchmarked
M8	Integrated prototype (text and audio) running in real time
M9	Full evaluation complete
M10	Final report, demonstration and deployment package

TABLE VI. PROJECT MILESTONES

13. Resources Required (Estimated Cost Calculation)

The college provides GPU facilities for training and ablation runs, and the team's existing laptops serve as developer machines, so the estimate covers only data-collection equipment and incidental expenses. All software used is open source.

Sr.	Item	Specification / Basis	Qty	Unit (₹)	Total (₹)
1	GPU / cloud compute	Institutional (college) GPU facility	—	0	0
2	Developer machines	Existing team laptops	4	0	0
3	Software and libraries	Python, TensorFlow/PyTorch, MediaPipe, OpenCV, gTTS (open source)	—	0	0
4	External webcam	1080p USB webcam for multi-camera recording	1	1,200	1,200
5	Tripod / mount	Fixed camera position and height variation	1	600	600
6	Portable LED light	Lighting-condition variation	1	700	700
7	Backup drive	256 GB USB 3.0 drive (landmark data is lightweight)	1	1,200	1,200
8	Refreshments and stationery	Signer recording sessions	Lump sum	—	1,000
9	Printing and binding	Proposal, report and poster	Lump sum	—	600
10	Local travel	Visits to the school and recording sites	Lump sum	—	700
	Estimated total				**≈ ₹6,000**

TABLE VII. ESTIMATED COST (INR)

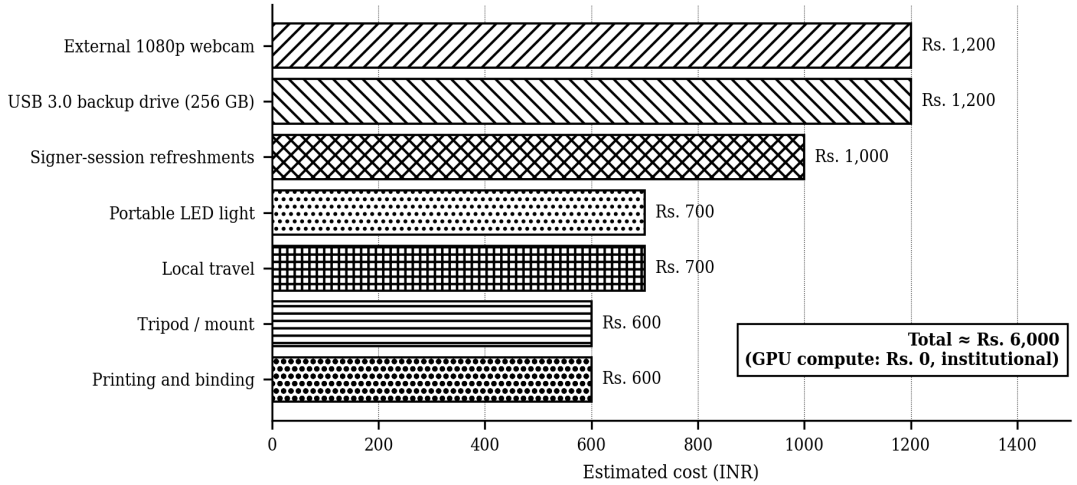


Fig. 6. Estimated cost breakdown by item (INR). *IEEE format – Black and White.*

14. References

[1] C. Lugaresi *et al.*, "MediaPipe: A framework for building perception pipelines," *arXiv preprint* arXiv:1906.08172, 2019.

[2] F. Zhang *et al.*, "MediaPipe Hands: On-device real-time hand tracking," *arXiv preprint* arXiv:2006.10214, 2020.

[3] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.

[4] M. Schuster and K. K. Paliwal, “Bidirectional recurrent neural networks,” *IEEE Trans. Signal Process.*, vol. 45, no. 11, pp. 2673–2681, Nov. 1997.

[5] A. Vaswani *et al.*, “Attention is all you need,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 5998–6008.

[6] C. Lea, M. D. Flynn, R. Vidal, A. Reiter, and G. D. Hager, “Temporal convolutional networks for action segmentation and detection,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 156–165.

[7] N. C. Camgoz, S. Hadfield, O. Koller, H. Ney, and R. Bowden, “Neural sign language translation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Salt Lake City, UT, USA, 2018, pp. 7784–7793.

[8] R. Rastgoo, K. Kiani, and S. Escalera, “Sign language recognition: A deep survey,” *Expert Syst. Appl.*, vol. 164, Art. no. 113794, Feb. 2021.

[9] A. Sridhar, R. G. Ganesan, P. Kumar, and M. Khapra, “INCLUDE: A large scale dataset for Indian Sign Language recognition,” in *Proc. 28th ACM Int. Conf. Multimedia*, 2020, pp. 1366–1375.

[10] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “BLEU: A method for automatic evaluation of machine translation,” in *Proc. 40th Annu. Meeting Assoc. Comput. Linguistics (ACL)*, Philadelphia, PA, USA, 2002, pp. 311–318.